

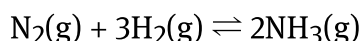
R2.3 How far? The extent of chemical change

Le Châtelier's principle

Learning outcomes

By the end of this section you should be able to apply Le Châtelier's principle to predict and explain responses to changes of systems at equilibrium.

The Haber process was first introduced in the Big Picture (see [section R2.3.0 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/)). The process is often described as 'making bread from air' because the nitrogen required for the reaction is obtained from the air and the ammonia produced is used to make fertiliser which in turn is used to grow crops for food production. The reaction itself is between nitrogen and hydrogen to produce ammonia.



During the Haber process, the conditions are carefully controlled to produce as much ammonia as possible, as economically as possible. These conditions include a very high pressure of approximately 20 000 kPa and a temperature of 450 °C. But why are such conditions necessary? The obvious answer would be to maximise the yield of ammonia. In this section, we see how the extent of a reaction can be influenced in order to maximise the yield of a chemical reaction.

Le Châtelier's principle is used to predict the effect on the equilibrium position when changes are made to a system at equilibrium. The equilibrium position can shift to the right or the left depending on the changes made to the reaction at equilibrium. If the equilibrium position shifts to the left, the concentration of reactants increases. Conversely if the equilibrium position shifts to the right, the concentration of products increases.

Le Châtelier's principle states that:

'When a system at equilibrium is subjected to a change, the system will respond to minimise the effect of the change.'

In its simplest form, it states that if changes are made to a system at equilibrium, the system will try to counteract the changes made. For example, if the temperature of a system at equilibrium is increased the system will try to decrease the temperature. Or if the pressure of a system at equilibrium is increased, the system will try to decrease the pressure.

The changes that can be made to a system at equilibrium are as follows:

1. Changes made to the concentration by either removing a reactant or product from the system at equilibrium.
2. Changes made to the pressure of a gaseous reaction by changing the volume of the reaction vessel, in accordance with Boyle's law (see [section S1.5.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ideal-gas-equation-id-44353/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ideal-gas-equation-id-44353/>)).
3. Changes made to the temperature of the system at equilibrium by either adding or removing heat.

Changes in concentration

First, we will consider changes in concentration. Again, we will use the reaction for the Haber process to explain the effect that changing the concentration has on the equilibrium position. **Figure 1** shows a graph of the concentrations of nitrogen, hydrogen and ammonia during the course of the reaction. The reaction is at equilibrium from 0 to 5 minutes as shown by the constant concentrations of the reactants and products. At five minutes, nitrogen is added to the reaction vessel which is shown on the graph by the vertical line. What effect does this have on the concentrations of hydrogen and ammonia? Study the graph and check your answer below when you are ready.

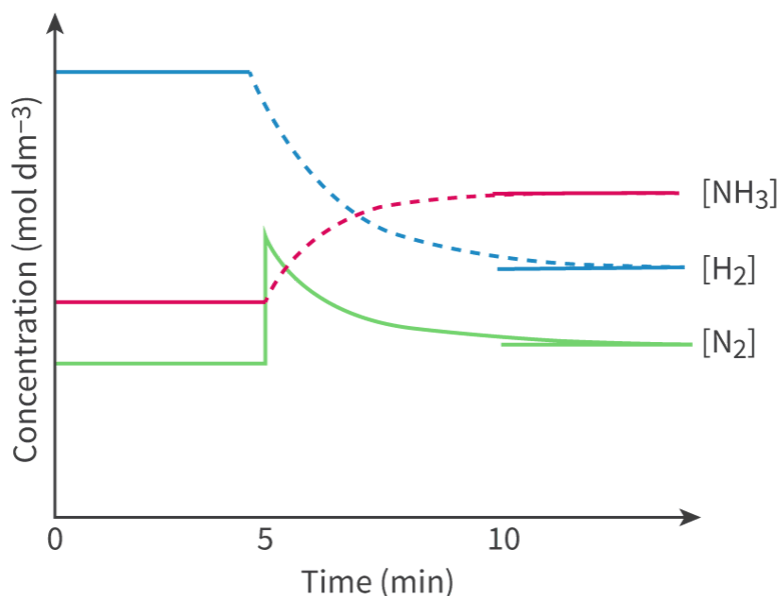


Figure 1. The effect of adding nitrogen on the concentrations of the reactants and products of the equilibrium mixture.

 More information for figure 1

This graph shows the concentrations of ammonia (NH_3), hydrogen (H_2), and nitrogen (N_2) over time during the Haber process reaction. The X-axis represents time in minutes ranging from 0 to 15 minutes, while the Y-axis represents concentration in mol dm^{-3} . Nitrogen concentration is indicated by a green line, which sharply increases at 5 minutes where a vertical line is shown. At this point, the hydrogen (H_2) concentration, shown by a blue dashed line, begins to gradually decrease, while the ammonia (NH_3) concentration, indicated by a pink dashed line, shows a slight constant increase after the 5-minute mark. The graph illustrates how adding nitrogen affects the equilibrium of this chemical reaction.

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As shown on the graph, the concentration of hydrogen decreases and the concentration of ammonia increases.

What has caused these changes in the concentrations of reactants and products? According to Le Châtelier's principle, when the nitrogen was added, the system responded by using up the additional nitrogen to produce more ammonia. This explains the increase in the concentration of ammonia that is observed. The concentration of the hydrogen decreased because it reacted with the added

nitrogen to produce ammonia. To summarise, adding nitrogen to the system at equilibrium shifted the equilibrium position to the right (the products side) which resulted in an increase in the yield of ammonia.

Removing a product will cause the equilibrium position to shift to the right, increasing the yield of the products(s).

The removal of a reactant will shift the equilibrium position to the left, increasing the yield of the reactant(s).

Changes in either the concentration of a reactant or a product have no effect on the value of the equilibrium constant.

Changes in pressure

Changes to the pressure of a system at equilibrium apply to gaseous reactions only. The pressure of a system can be changed by changing the volume of the reaction vessel according to Boyle's law (**Figure 2**). A decrease in the volume of the reaction vessel will increase the pressure whereas an increase in the volume will decrease the pressure.

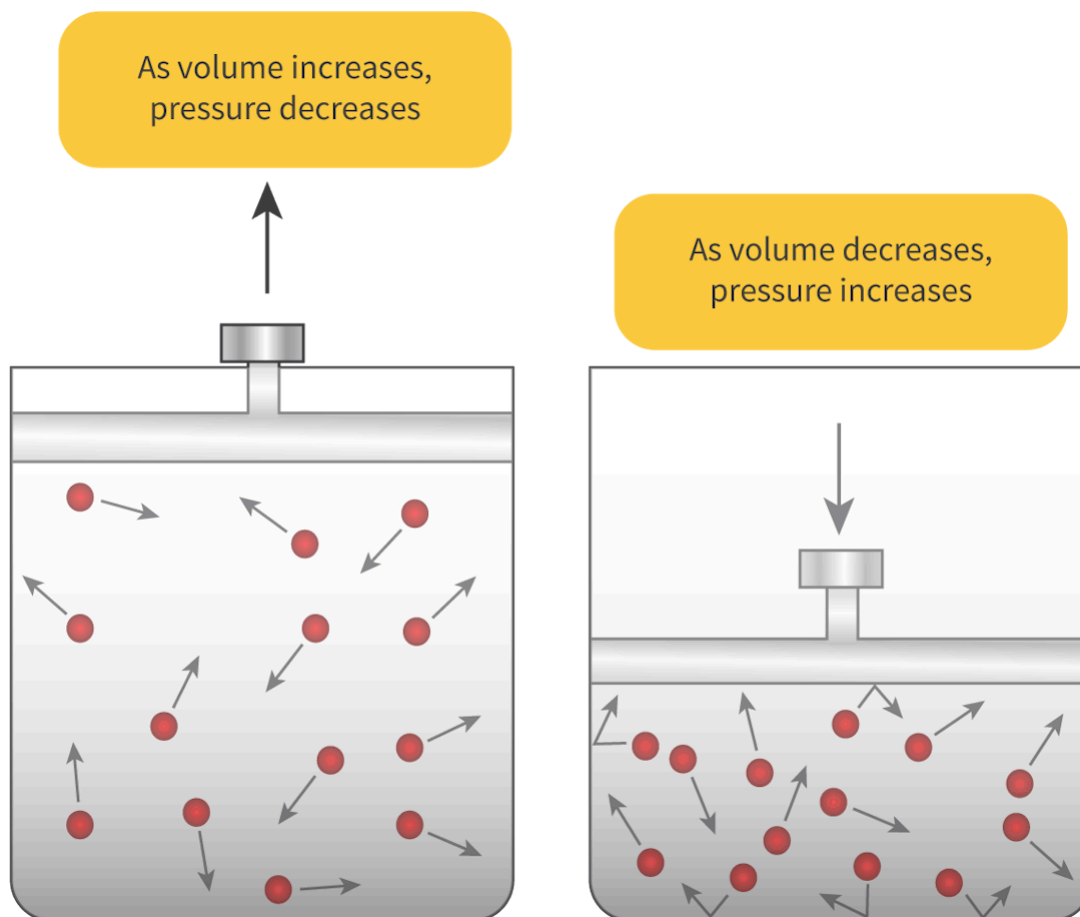


Figure 2. Boyle's law states that pressure and volume are inversely proportional at constant temperature.

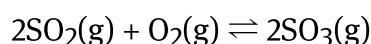
 More information for figure 2

The image depicts Boyle's law with two separate diagrams of a piston. On the left, the piston is raised, allowing for an increase in volume. Red spheres representing gas molecules are spread out with arrows indicating their movement. A yellow box above the diagram reads, 'As volume increases, pressure decreases,' illustrating the inverse relationship between volume and pressure at constant temperature.

On the right, the piston is lowered, resulting in a decrease in volume. The gas molecules are closer together, and their movement is denoted by arrows. The corresponding yellow box states, 'As volume decreases, pressure increases,' reinforcing Boyle's law principle. The illustration uses visual positioning and arrows to communicate the change in molecular movement corresponding to changes in volume and pressure.

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To determine which side the equilibrium position will shift to when the pressure of the system is changed, count the number of moles of gas on each side of the equation. This can be done by simply adding up the coefficients of the reactants and products in the balanced equation. For example, in the reaction between sulfur dioxide and oxygen to produce sulfur trioxide, we can see that there are a total of three moles of gas in the reactants and two moles of gas in the products.



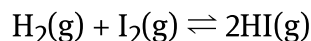
Once the number of moles of gas in the reactants and products have been determined, the shift in the equilibrium position can be predicted as follows:

- An increase in pressure favours the side with the fewer number of gaseous molecules. This occurs because at high pressure, gas particles will experience a greater frequency of collisions, increasing their rate of reaction.
- A decrease in pressure favours the side with the greater number of gaseous molecules. This occurs because at low pressure, the

particles are farther apart and have a lower frequency of collision, therefore a decrease in the rate of reaction.

So for the reaction shown above, what is the effect of changing the pressure of the system at equilibrium? If the pressure is increased, the equilibrium position shifts to the right because the product side has a fewer number of gaseous molecules. This has the effect of decreasing the pressure of the system. And conversely, if the pressure is decreased, the equilibrium position shifts to the left because the reactants side has a greater number of gaseous molecules. This has the effect of increasing the pressure of the system.

It is important to note that changes in pressure only have an effect on the equilibrium position when there are different numbers of gaseous molecules in the reactants and products. Take for example, the reaction between hydrogen and iodine to produce hydrogen iodide.



If the pressure of this system changes, there will be no effect on the equilibrium position because there are two moles of gas on each side of the equation.

Changes in pressure have no effect on the value of the equilibrium constant because the equilibrium constant only varies with a change in temperature.

Changes in temperature

To predict the effect of a change in temperature on the equilibrium position we need to know the enthalpy change, ΔH , for the forward (or reverse) reaction. A negative enthalpy change means that the forward reaction is exothermic [section R1.1.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/endothermic-and-exothermic-reactions-id-45282/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/endothermic-and-exothermic-reactions-id-45282/) with heat released from the system to the surroundings and the reverse reaction is endothermic with heat being absorbed from the surroundings to the system (**Figure 3**).

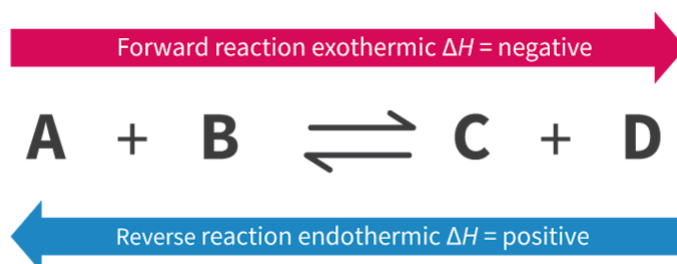


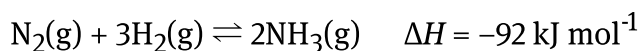
Figure 3. The enthalpy change of the forward reaction is the opposite of the reverse reaction.

 More information for figure 3

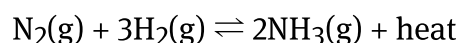
The diagram illustrates two opposing chemical reactions with their respective enthalpy changes. At the center, two reactants, labeled 'A' and 'B', are shown reacting to form two products, 'C' and 'D'. There is a double-headed arrow between the reactants and products, indicating the reversible nature of the reaction. Above this reaction, a pink arrow points to the right, labeled 'Forward reaction exothermic $\Delta H = \text{negative}$ ', indicating that the forward reaction releases heat to the surroundings and has a negative enthalpy change. Below the reaction, a blue arrow points to the left with the label 'Reverse reaction endothermic $\Delta H = \text{positive}$ ', signifying that the reverse reaction absorbs heat from the surroundings and has a positive enthalpy change.

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Returning to the reaction for the Haber process, we see it has a negative enthalpy change for the forward reaction.

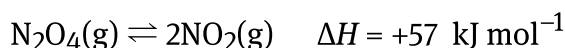


In this context, it is helpful to consider heat as either a reactant for an endothermic reaction or as a product for an exothermic reaction. So we can rewrite the equation for the Haber process as:



If heat is added to the system at equilibrium it can be thought of as the addition of a product which shifts the equilibrium position to the left in the endothermic direction and if heat is removed from the system at equilibrium, the equilibrium position will shift to right in the exothermic direction.

Next we will consider the effect of adding or removing heat on an endothermic reaction.



This equation can be rewritten to include the heat as a reactant:



Now we can see that adding heat to the system at equilibrium will shift the equilibrium position to the right in the endothermic direction and removing heat will shift the equilibrium position to the left in the exothermic direction.

Temperature and the equilibrium constant

Now we will look at the effect of changing the temperature on the value of the equilibrium constant. Recall that the equilibrium constant is temperature dependent meaning that its value changes if the temperature of the reaction is changed. If the equilibrium position shifts to the right, the value of the equilibrium constant will increase and if the equilibrium position shifts to the left, the value of the equilibrium constant decreases. So for an exothermic reaction, increasing the temperature shifts the equilibrium position to the left and the value of the equilibrium constant decreases and for an endothermic reaction, increasing the temperature shifts the equilibrium position to the right and the value of the equilibrium constant increases.

Try **Interactive 1** to test your understanding of temperature and equilibrium.

	Exothermic reaction	Endothermic reaction
Effect on equilibrium position when temperature increases		
Effect on equilibrium position when temperature decreases		
Effect on K_c when temperature increases		
Effect on K_c when temperature decreases		

Shifts to the left

Shifts to the right

 K_c decreases K_c increases

✓ Check

Interactive 1. Temperature and equilibrium.

Catalysts and equilibrium

You learned previously in [section R2.2.4–5](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/>) that a catalyst increases the rate of a chemical reaction by providing an alternative reaction pathway with a lower activation energy. A catalyst added to a reversible reaction will allow the reaction to reach an equilibrium position sooner but has no effect on the equilibrium position or on the value of the equilibrium constant. This is because the rate of the forward reaction and the rate of the reverse reactions are increased by the same amount.

Making connections

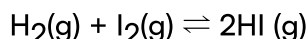
Why do catalysts have no effect on the value of the equilibrium constant or on the equilibrium composition subtopic R2.2

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43127/>)?

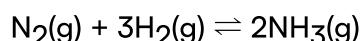
Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

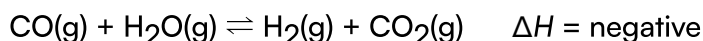
1. Predict the effect on the yield of hydrogen iodide and on the value of the equilibrium constant when the concentration of hydrogen is increased in the following reaction.



2. Predict the effect on the equilibrium position, the yield of ammonia and on the value of the equilibrium constant when the volume of the reaction vessel is decreased at constant temperature for the following reaction.



3. Predict the effect on the equilibrium position, the yield of hydrogen and on the value of the equilibrium constant when the temperature of the system is increased for the following reaction.



4. Explain the effect of adding a catalyst to a reaction at equilibrium.

1. Adding hydrogen will shift the equilibrium position to the right which will increase the yield of the hydrogen iodide. The value of the equilibrium constant will remain unchanged.
2. Decreasing the volume of the reaction vessel at constant temperature will increase the pressure of the system. The equilibrium position will shift to the side that has the fewer number

of gaseous molecules, which is the products side. The yield of ammonia will increase. The value of the equilibrium constant will remain unchanged.

3. Increasing the temperature for an exothermic reaction will shift the equilibrium position to the left in the endothermic direction. The yield of hydrogen will decrease as will the value of the equilibrium constant.
4. A catalyst has no effect on the equilibrium position because it increases the rate of the forward and reverse reactions by the same amount.